Albert Gafiyatullin

AI Compiler Engineer

Email: albert.gafiyatullin@outlook.com
Github: a-gafiyatullin
LinkedIn: Albert G.

Summary

Software Engineer specializing in the development of programming language compilers and runtimes, with experience in AI compiler optimizations, JIT compilation, and JVM internals.

Experience

- **Samsung Research Russia** — Moscow, Russia
  Leading AI Compiler Engineer, SOC SW Lab — *July 2023 - Present*
  C++ | Compilers | Neural Networks | Optimizations

  Developing the optimizing AI compiler for the Samsung Exynos SoC NPU:

  - Designed and implemented new generic SPM-allocation, post-allocation, and scheduling optimizations in the NPU compiler, improving neural-network performance and energy efficiency on Samsung mobile devices;
  - Reduced inference latency for target neural networks through performance analysis and tuning.

- **Unipro** — Novosibirsk, Russia
  JIT Compiler Engineer, Java Virtual Machine Team — *March 2021 - June 2023*
  C++ | JVM | Assembly | Compilers | Garbage Collection

  JVM runtime and JIT compiler development for MCST's Elbrus VLIW processor:

  - Adapted the C2 compiler to support tiered compilation, improving average startup performance by 50% compared with non-tiered compilation;
  - Designed and implemented a fast "C0" compiler for warm-up compilation tiers, replacing C1 and targeting the Elbrus VLIW architecture; improved average startup performance by 25% on low-core-count systems compared with C2-based tiered compilation;
  - Improved performance of string- and XML-related workloads by up to 8% using intrinsics;
  - Reduced runtime overhead by implementing platform-specific improvements to implicit null checks.

Projects & Courses

- **COOL Compiler** — *March 2024*
  C++ | LLVM | Garbage Collection | Compilers | ARM

  Implemented a COOL compiler and runtime using LLVM:

  - Supported AArch64 and x86-64 target architectures;
  - Implemented stop-the-world mark-and-sweep, mark-and-compact and semispace copying garbage collectors.

- **SOE.YCSCS1: Compilers** — *October 2021*
  C++ | MIPS | Compilers | Assembly

  Implemented a COOL compiler for the SPIM emulator.

- **ENGR85A: Digital Design** — *April 2023*
  SystemVerilog | Circuit Design

  Combinational and sequential circuit design.

- **ENGR85B: Computer Architecture** — *June 2023*
  RISC-V | Computer Architecture | SystemVerilog | Embedded Systems | Circuit Design

  Implemented a multicycle RISC-V CPU and studied introductory pipelined CPU design.

Education

- **Novosibirsk State University** — Novosibirsk, Russia
  Master's degree in Computer Science — *2021 - 2023*
  - **Thesis**: Tiered JIT Compilation in the Java Virtual Machine for the Elbrus Platform.
- **Novosibirsk State University** — Novosibirsk, Russia
  Bachelor's degree in Computer Science — *2017 - 2021*
  - **Thesis**: Development of a computational module for simulating the destruction of a fast-neutron reactor core.
  - **GPA**: 4.8/5.0, graduated with honors

Programming Skills

- **Languages:** C++, C, assembly, Python.
- **Technologies:** JVM internals, compiler optimization, JIT compilation, computer architecture, LLVM, garbage collection